

John (“Jack”) Klawitter

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EDUCATION

Rutgers University, New Brunswick, NJ

Doctor of Philosophy (Ph.D.) in Computer Science (Enrolled)

- Computational Brain Lab, CBIM
- Advisor: Dr. Konstantinos Michmizos

Middlebury College, Middlebury, VT

Bachelor of Arts with Majors in Physics and Computer Science, Magna Cum Laude – February 2023

RESEARCH and WORK EXPERIENCE

Graduate Research Assistant, Rutgers Computer Science Department **Fall 2023-Present**

- Developed interpretable machine learning models for decoding high-dimensional neural recordings, enabling analysis of representational drift across motor learning
- Introduced Critical Spike Attribution, a novel feature attribution method for spiking neural networks, providing causal insight into spike-level contributions to model predictions
- Applied representation analysis and feature attribution to align model dynamics with biological change
- Led collaboration between Computer Science and Psychology labs to integrate modeling and experimental data collection

Research Assistant, Middlebury Physics Department **Spring and Fall 2022**

- Developed Python script to correct the spectra of red quasars
- Published open-source code for use by the astrophysics community

Machine Learning Intern, Rolls-Royce **Summer 2022**

- Developed time-series deep learning models for early detection of jet engine failures, improving detection rate by approximately 10% in operational data

Undergraduate Researcher, Ithaca College Dynamical Systems REU **Summer 2021**

- Studied billiard dynamics on surfaces of revolution
- Developed a Mathematica notebook that models billiard dynamics on any surface of revolution within a “billiard board” — any region bounded by a piecewise-defined curve
- Developed original billiard constructions, including a new “unilluminable room” on the sphere and a novel result regarding the existence of periodic orbits on surfaces of revolution.
- Presented findings at the 2022 Joint Mathematics Meeting poster session

Research Assistant, Middlebury Math Department **Summer 2020**

- Worked on Martin Gardner’s “Minimum No-3-In-A-Line” Problem
- Examined the use of SAT Solvers as they apply to combinatorial problems
- Modified programs written by Dr. Donald Knuth and created novel Python scripts

PUBLICATIONS

- **Klawitter & Michmizos**, “Critical Spike Attribution for Feature Importance in Spiking Neural Networks”, *Neuro Inspired Computational Elements*, 2026 ([Oral Presentation](#))
- **Klawitter & Michmizos**, “[Sequential Learning in Neural Decoders Reveals Interpretable Neural Representations](#)”, *Neurocomputing*, 2026. 132742, ISSN 0925-2312.

AWARDS/HONORS

- Spring 2026 Rutgers Brain Health Institute Trainee Travel Award

TEACHING

Course Instructor: Design and Analysis of Computer Algorithms (Summer 2024)

Teaching Assistant: Intro to Discrete Structures I (Spring 2024, Spring 2025), Intro to Discrete Structures II (Fall 2024), Intro to Artificial Intelligence (Summer 2025)

SKILLS

Programming: Python (NumPy, PyTorch, scikit-learn)

Machine Learning: Deep learning, spiking neural networks, neural decoding, time-series modeling, representation analysis, feature attribution, interpretable ML

Tools: Git, LaTeX